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Ethan Winn Bishop

Provo, UT | Open to Relocation

WORK EXPERIENCE

VFX/Technical Artist - BYU Animation — AUGUST 2024 - PRESENT

- Led a team of 3 VFX artists to create realtime VFX and tools for capstone game ‘Dragonkisser’
- Used Unreal Widget Blueprints, Blueprints, C++, and Niagara to create environment, FX, and UI tools within Unreal and Houdini to allow artist control while maintaining instances and shader optimizations
- Worked in a team of 4 FX artists to develop FX and rendering tools in Houdini for capstone short film ‘Honey Business’ including honey, grass, bee swarms, and USD tools for instanced variants

Teaching/Research Assistant - BYU Animation — AUGUST 2024 - MAY 2025

- Assisted 3 classes of 20-30 students in understanding programming, procedural modeling, rendering, FX, and shading in Houdini and Maya through one-on-one and group sessions
- Worked in a team of 8 FX artists to develop FX tools within Houdini for capstone short film ‘Love & Gold’ including torch fires, dragon fire blasts, coin RBD, and various magic FX

Motion Graphics Technical Artist - BYU Broadcasting — SEPTEMBER 2022 - JANUARY 2025

- Trained 6 new hires in C4D, Blender, After Effects
- Scripted procedural animation and template tools within C4D, Blender, Houdini, and After Effects
- Assisted setting up a render farm network using Deadline

PROJECTS

Realtime GPU Pathtracer

- Used the Vulkan API within a C++ framework to create a realtime pathtracer capable of 60 FPS at 1080p with over 250,000 triangles on RTX 4070 Ti hardware with animated lights and geometry
- Create the shaders in GLSL (raygen, PBR materials, cook-torrence BRDF, and denoising)
- Scripted a bash tool to automatically convert shaders to slang at build time

Digging Mini Game

- Created a custom Skidsteer vehicle in UE5 and C++ that can add and remove to/from a custom C++ voxel terrain class
- Coded a custom script in Houdini that prepares and exports VDBs for use in voxel terrain generation
- Used UE5, C++, and Houdini to create a custom vehicle system based on UE5 Chaos Vehicle for specific movement and collision with the unusual terrain

EDUCATION

Brigham Young University - Provo, UT — SEPTEMBER 2021 - APRIL 2026
BS in Computer Science Animation | GPA: 3.89

SKILLS

Programming - C++, C, Python, VEX, GLSL, HLSL, Java, JavaScript, OpenMP
3D Software - Houdini, Maya, Cinema4D, Blender (including geometry nodes), EmberGen
Game Engines - Unreal Engine (including Material graph, Blueprints, Niagara), Godot, Unity
Shading & Rendering - Substance Painter, Substance Designer, RenderMan, Redshift, Arnold
Tools & Technologies - OpenGL, Vulkan, Git, Perforce, NVIDIA Nsight, Unreal Insight, CMake, Solaris, USD, Linux

REFERENCES

Seth Holladay 801-422-6490 holladay@cs.byu.edu | Parris Egbert 801-422-4029 egbert@cs.byu.edu